

Week 12

Jenkins-Docker Continuous Deployment

FINAL VALIDATION | JENKINS BUILD #20 | SUCCESS

This report records the final successful end-to-end Week 12 deployment. Jenkins checked out the verified source revision, passed the Selenium regression gate, built and published a versioned Docker image, deployed a fresh application container, and confirmed the live login endpoint returned HTTP 200.

Successful Build at a Glance

Jenkins job / build	Community-Health-Survey-Pipeline / #20
Result and duration	SUCCESS / 3 minutes 41 seconds
Source revision	552b3c2cdd3837c94118560b8efdb69af51c9a01
Selenium regression gate	9 tests • 0 failures • 0 errors • 0 skipped
Published Docker image	jyotsnakasi/chss-app:build-20
Deployed endpoint	HTTP 200 • http://localhost:18082/login

Week 12 Objective

Automate continuous deployment from Jenkins to Docker Hub and a fresh Docker container. The pipeline must publish a uniquely tagged image only after tests pass, verify the registry artifact, replace the CHSS container, and fail if container or application health checks do not pass.

Jenkinsfile and Pipeline Implementation

The Jenkinsfile keeps the regression gate ahead of every package and deployment action. JUnit XML and Selenium failure artifacts are published, and a non-zero Maven test result prevents the pipeline from reaching Tomcat or Docker deployment.

Pipeline stage	What it does
Checkout	Checks out the selected Git revision. Build #20 used the revision shown in the evidence.
Build	Runs Maven clean/test-compile before the browser suite.
Selenium Regression Gate	Runs the Selenium suite against the isolated test app. Build #20 passed 9/9.
Publish Reports	Publishes JUnit XML and archives Selenium failure screenshots when present.
Package	Creates the Community Health Survey System WAR after the gate passes.
Tomcat Deployment	Deploys the WAR and verifies the Tomcat application endpoint.
Docker Build	Builds the application image tagged with the Jenkins build number.
Docker Hub Login	Binds Jenkins credentials and sends the token through docker login --password-stdin.
Docker Push	Publishes the versioned image to the authenticated Docker Hub repository.
Registry Verification	Inspects the remote Docker Hub manifest for the pushed build tag.
Fresh Container Deployment	Pulls the exact image and launches chss-container with host port 18082 mapped to 8081.
Container + Health Checks	Verifies image, running state, port mapping, startup logs, and HTTP 2xx health.

Gate behavior: Selenium failure → report publication → pipeline failure; Package, Tomcat, Docker Hub, and container deployment stages remain blocked.

Successful Jenkins Build #20

The Jenkins build page records a green SUCCESS result, the exact source revision, test reports, and the Week 12 status-update regression commit. The published reports correspond to the 9-test Selenium/JUnit suite: 9 passed, 0 failures, 0 errors, 0 skipped.

Job and build	Community-Health-Survey-Pipeline / #20
Revision	552b3c2cdd3837c94118560b8efdb69af51c9a01
Build result	SUCCESS • 3m 41s
Selenium result	9 passed • 0 failures • 0 errors • 0 skipped

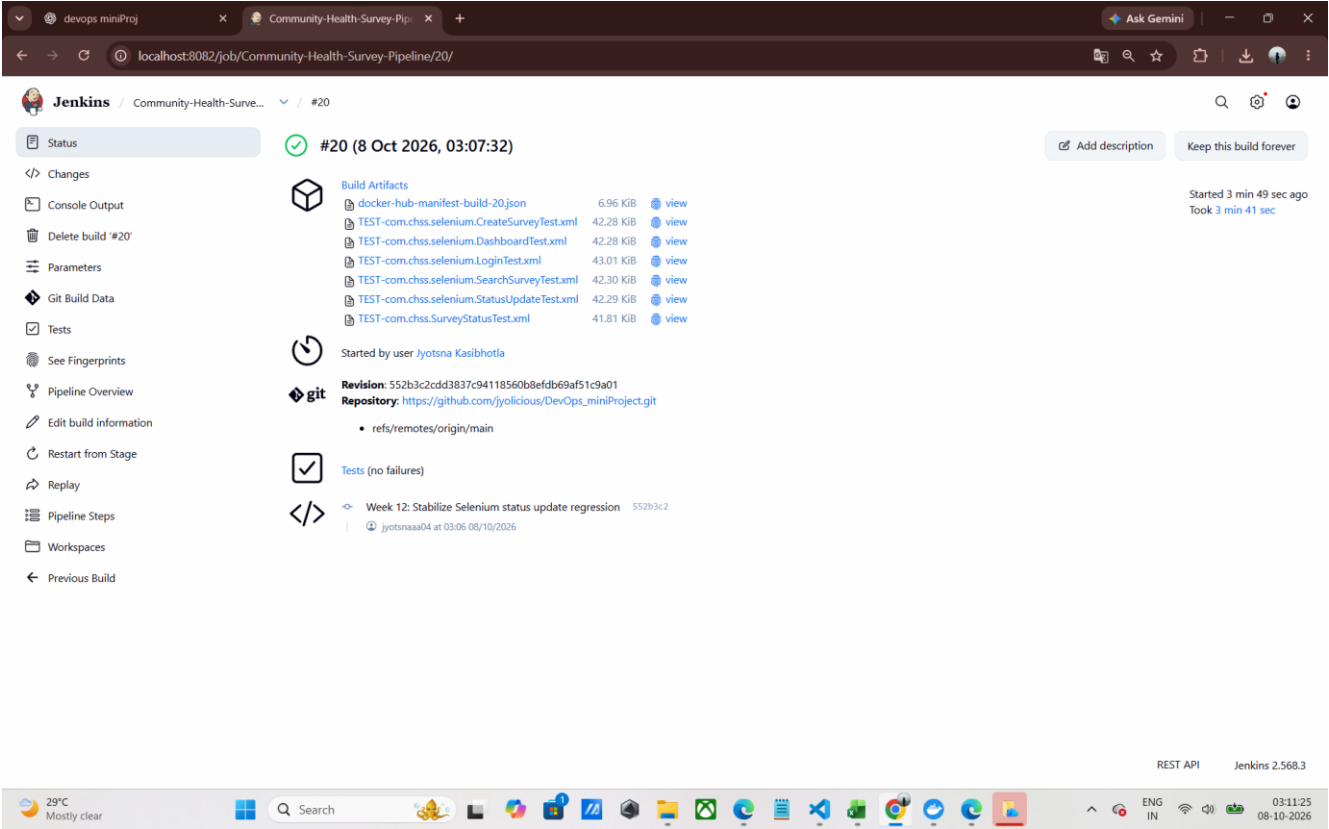


Figure 1. Actual Jenkins Build #20 status page supplied for this report; revision, successful result, and JUnit artifacts are visible.

The pipeline’s final success message was: “Community Health Survey System deployed successfully!”

Docker Hub Registry Evidence

The versioned image produced by Build #20 was pushed to Docker Hub and its remote manifest was verified by the pipeline. The supplied Docker Hub page shows the published tag and its image metadata.

Repository and tag	jjyotstnakasi/chss-app:build-20
Platform	linux/amd64
Compressed size	134.27 MB
Registry result	Docker Hub push succeeded; manifest verification succeeded.

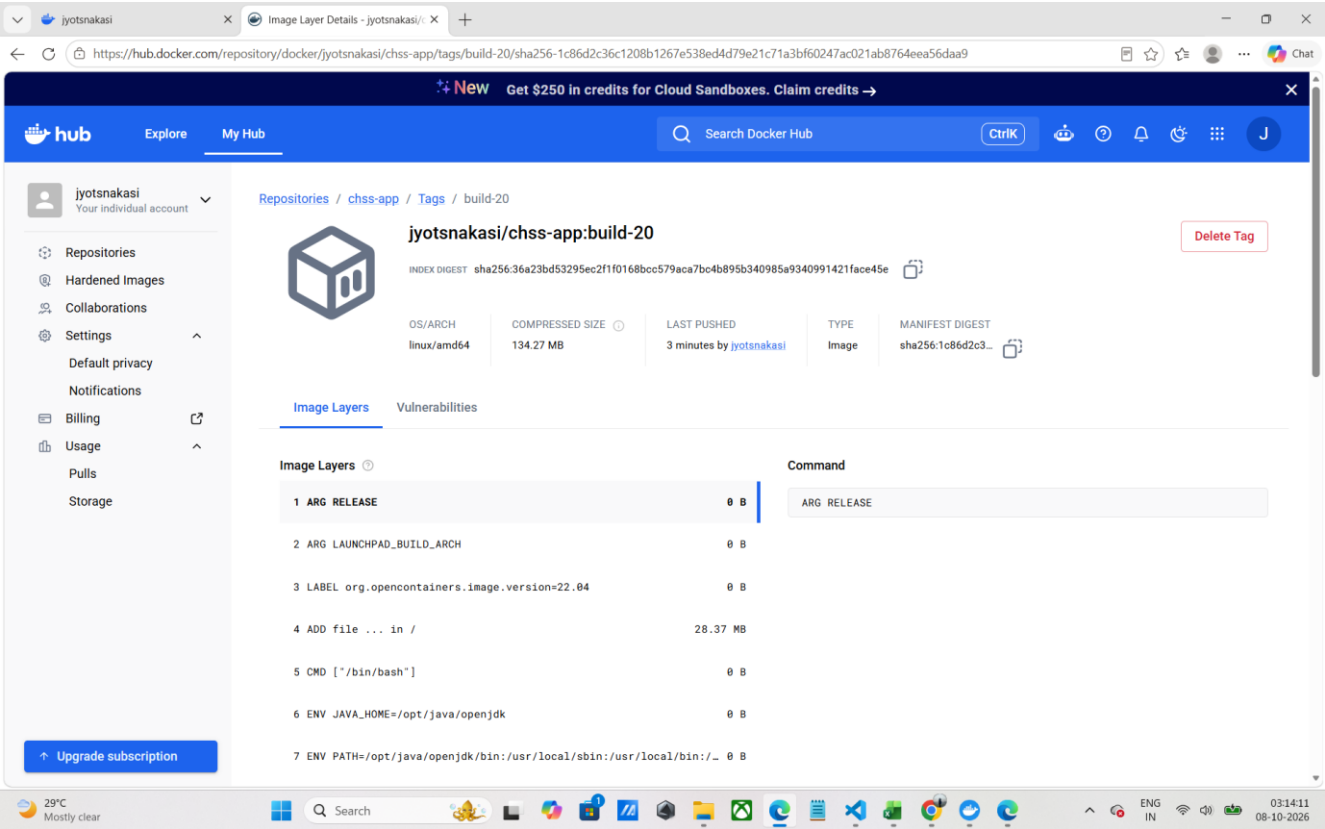


Figure 2. Actual Docker Hub build-20 tag page supplied for this report. The repository/tag, platform, compressed size, last-pushed information, and image layers are visible.

The screenshot is included as supplied. No credentials are shown, and no digest is transcribed into the report.

Fresh Container Deployment and Health Verification

After registry verification, Jenkins pulled the exact build-20 image and deployed it as a fresh CHSS container. Container verification checked the image reference, running state, application startup logs, and published port mapping.

Container name	chss-container
Image	jjyotsnakasi/chss-app:build-20
Port mapping	Host 18082 → container 8081
Container verification	Passed: expected image, running container, and 18082:8081 mapping.
Application health	HTTP 200 • http://localhost:18082/login

End-to-End Result

Git revision → Jenkins checkout → Maven build → Selenium 9/9
→ WAR packaging and Tomcat deployment → Docker image build
→ Docker Hub login and push → registry manifest verification
→ fresh chss-container → container verification → HTTP 200
→ **Jenkins SUCCESS**

Security and Conclusion

Docker Hub authentication remains in Jenkins credentials and is passed to Docker via `--password-stdin`. The Docker Hub PAT and database password are not included in this report or its screenshots.

Week 12 successfully demonstrates Jenkins-Docker continuous deployment: after the Selenium gate passed, Jenkins published the versioned Docker Hub image, deployed a fresh container, and verified the running application endpoint before reporting SUCCESS.